Machine Learning

02-06-26

Activation	hidden layers	output layer
Linear	✓	✓
sigmoid	✗	✓
relu	✓	✗
Leaky relu	✓	✗
ELU	✓	✗
softmax	✗	✓
tanh	✓	✗

ANN Regression

eg:	x_1	x_2	y
	:	:	0.2
	:	:	0.3
	:	:	0.4
			:

ANN → 2 layers = 1 hidden layer + 1 output layer

- 1 hidden layer → 2 neurons → ReLU activation
- 1 output layer → 1 neuron → Linear activation

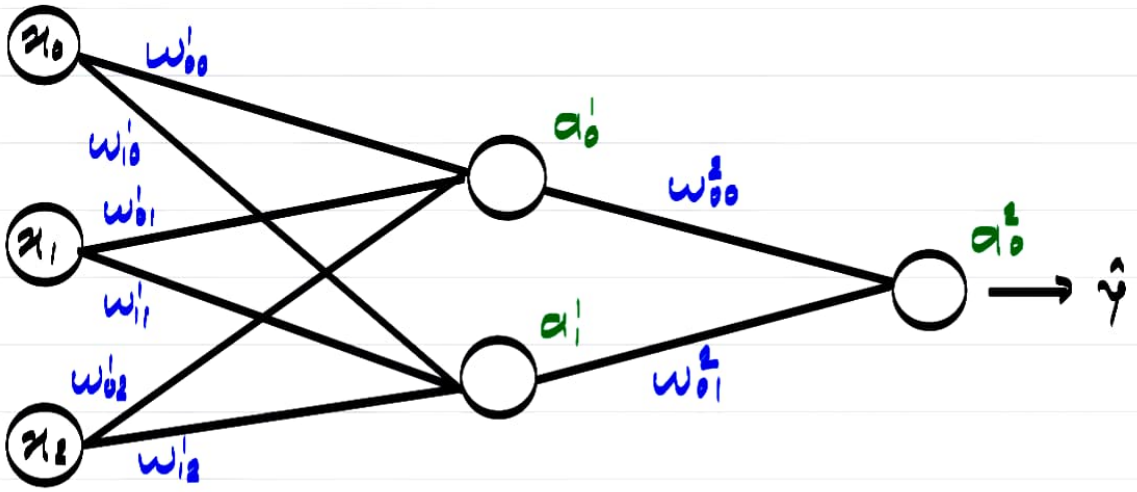
In ANN regression, output layer always has single neuron.

x_1	x_2	x_0	y
		1	0.2
		1	0.3
		1	0.4

$z = w_1 x_1 + w_2 x_2 + b$

$z = w_1 x_1 + w_2 x_2 + w_0 x_0$

to accomodate bias in weight matrix



input layer	hidden layer	output layer
Layer 0	Layer 1	Layer 2

Forward Propagation :

Data matrix $X = \begin{bmatrix} x_{10} & x_{20} & x_{30} & \dots & x_{n0} \\ x_{11} & x_{21} & x_{31} & & x_{n1} \\ x_{12} & x_{22} & x_{32} & & x_{n2} \end{bmatrix}_{3 \times n}$

$\downarrow$ d_1 $\quad$ $\downarrow$ d_2 $\quad$ $\downarrow$ d_3 $\quad$ $\downarrow$ d_4

total 'n' training datapoints

Layer 1 :

$$z^1 = w^1 x$$

$$z^1 = \begin{bmatrix} w^1_{00} & w^1_{01} & w^1_{02} \\ w^1_{10} & w^1_{11} & w^1_{12} \end{bmatrix} \begin{bmatrix} x_{10} & x_{20} & x_{30} & \dots & x_{n0} \\ x_{11} & x_{21} & x_{31} & & x_{n1} \\ x_{12} & x_{22} & x_{32} & & x_{n2} \end{bmatrix}$$

2×3 $\quad$ $3 \times n$

$$z^1 = \begin{bmatrix} z^1_{01} & z^1_{02} & z^1_{03} & \dots & z^1_{0n} \\ z^1_{11} & z^1_{12} & z^1_{13} & \dots & z^1_{1n} \end{bmatrix}$$

$$a^1 = \text{ReLU}(z^1)$$

$$= \begin{bmatrix} \text{relu}(z^1_{01}) & \text{relu}(z^1_{02}) & \text{relu}(z^1_{03}) & \dots & \text{relu}(z^1_{0n}) \\ \text{relu}(z^1_{11}) & \text{relu}(z^1_{12}) & \text{relu}(z^1_{13}) & \dots & \text{relu}(z^1_{1n}) \end{bmatrix}$$

$$a^1 = \begin{bmatrix} a^1_{01} & a^1_{02} & a^1_{03} & \dots & a^1_{0n} \\ a^1_{11} & a^1_{12} & a^1_{13} & & a^1_{1n} \end{bmatrix}$$

a^1_{01}, a^1_{11} } 2 outputs generated by 2 nodes in layer 1 for datapoint d_1

such 2 outputs are there for all 'n' datapoints in a^1 matrix

w^1 → weight matrix in layer 1

w^1 shape = 2 rows × 3 cols

- 2 rows → no. of nodes in layer 1 (current layer)
- 3 cols → no. of nodes in layer 0 (previous layer)

for layer l

W^l shape: no. of nodes in layer l $\times$ no. of nodes in layer $l-1$

Layer 2:

$$Z^2 = W^2 A^1$$

$$= \underset{1\times 2}{\begin{bmatrix} w^2_{10} & w^2_{01} \end{bmatrix}} \underset{2\times n}{\begin{bmatrix} a^1_{01} & a^1_{02} & a^1_{03} & \dots & a^1_{0n} \\ a^1_{11} & a^1_{12} & a^1_{13} & & a^1_{1n} \end{bmatrix}}$$

$$Z^2 = \underset{1\times n}{\begin{bmatrix} z^2_{01} & z^2_{02} & z^2_{03} & \dots & z^2_{0n} \end{bmatrix}}$$

A^2: Linear Act (Z^2)

$$A^2 = \underset{1\times n}{\begin{bmatrix} a^2_{01} & a^2_{02} & a^2_{03} & \dots & a^2_{0n} \end{bmatrix}}$$

$a^2_{01} \to d_1$, $a^2_{02} \to d_2$, $a^2_{03} \to d_3$, $a^2_{0n} \to d_n$

$$\hat{Y} = A^2$$

- As this is regression, loss function used will be MSE
- We'll minimize MSE by finding optimal weights using gradient descent.

$$Y = [\, y_1 \;\; y_2 \;\; y_3 \dots y_n \,] \rightarrow \text{Yact}$$

$$\hat{Y} = [\, \hat{y}_1 \;\; \hat{y}_2 \;\; \hat{y}_3 \dots \hat{y}_n \,] = [\, a^2_{01} \;\; a^2_{02} \;\; a^2_{03} \dots a^2_{0n} \,]$$

$\hookrightarrow$ Ypred

Backward Propagation:

$d_1 \rightarrow x_{01}\ \ x_{11}\ \ x_{12} \qquad y_1$

$$z^1_{01} = w^1_{00} x_{01} + w^1_{01} x_{11} + w^1_{02} x_{12}$$

$$a^1_{01} = relu(z^1_{01})$$

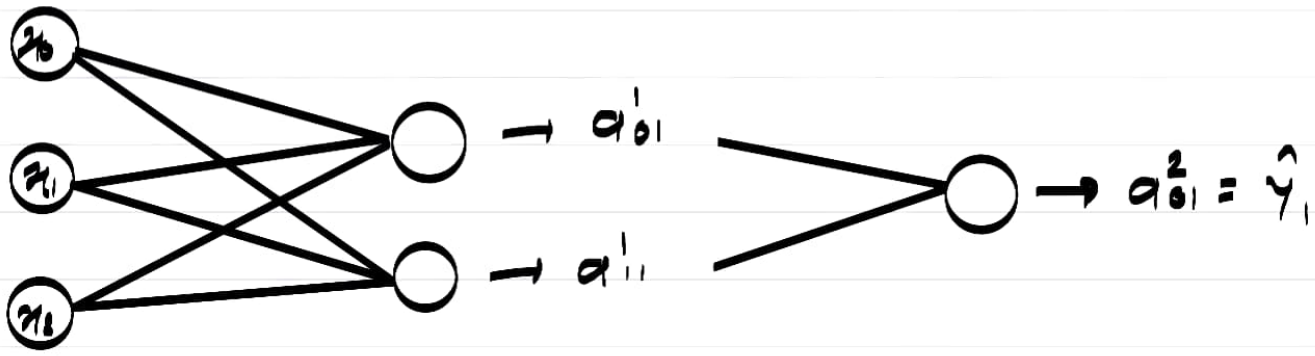
↳ output of node 0 in layer 1 for d_1

$$z^1_{11} = w^1_{10} x_{01} + w^1_{11} x_{11} + w^1_{12} x_{12}$$

$$a^1_{11} = relu(z^1_{11})$$

↳ output of node 1 in layer 1 for d_1

d_1



$$z^2_{01} = w^2_{00} a^1_{01} + w^2_{01} a^1_{11}$$

$$a^2_{01} = Linear(z^2_{01})$$

$$\hat{y}_1 = a^2_{01}$$

Loss MSE: $L = (y_1 - \hat{y}_1)^2$

$$L = (y_1 - \hat{y}_1)^2$$

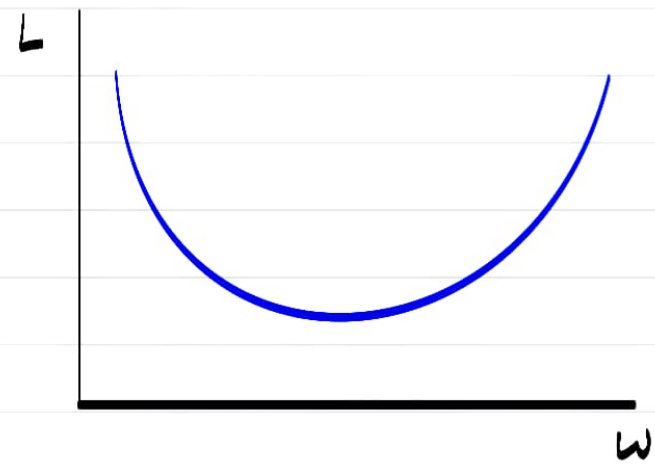
$$= (y_1 - a^2_{01})^2$$

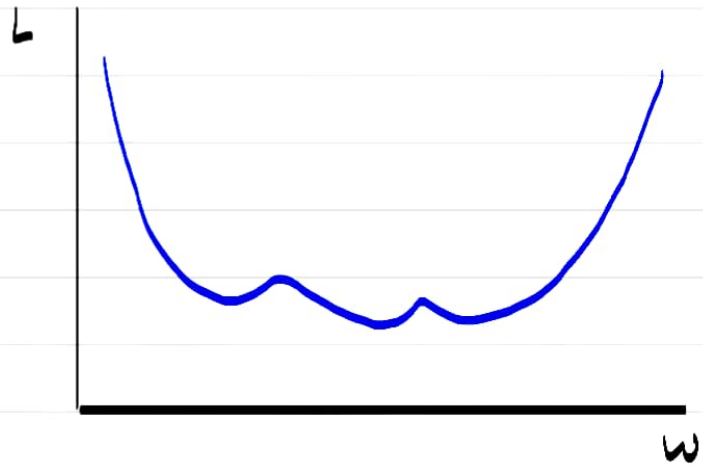
$$= [y_1 - \text{Linact}(z^2_{01})]^2$$

$$L = [y_1 - \text{Linact}(w^2_{00}\, a^1_{01} + w^2_{01}\, a^1_{11})]^2$$

- we need to update w to minimize loss

- We can use gradient descent here as relⁿ between loss and weights is convex.





$$w_{new} = w_{old} - \alpha\, \partial L / \partial w_{old}$$

we need $\partial L / \partial w$ to update w

lets pick w^2_{01} from layer 2

for dp loss is

$$L = (y_1 - \hat{y}_1)^2 = (y_1 - a^2_{01})^2 \longrightarrow \partial L / \partial a^2_{01}$$

$$\hat{y}_1 = a^2_{01}$$

we need to find $\partial L / \partial w^2_{01}$

$$a^2_{01} = \text{Lin act}(z^2_{01}) \longrightarrow \partial a^2_{01} / \partial z^2_{01}$$

$$z^2_{01} = w^2_{00} a^1_{01} + w^2_{01} a^1_{11} \longrightarrow \partial z^2_{01} / \partial w^2_{01}$$

$$\frac{\partial L}{\partial w^2_{01}} = \frac{\partial L}{\partial a^2_{01}} \times \frac{\partial a^2_{01}}{\partial z^2_{01}} \times \frac{\partial z^2_{01}}{\partial w^2_{01}} \longrightarrow \text{using chain rule}$$

this is for datapoint dp

$$L = (y_1 - \hat{y}_1)^2$$

$$L = (y_1 - a^2_{01})^2$$

lets put $(y_1 - a^2_{01}) = t$

$L = t^2 \quad \partial L / \partial t = 2t$

$t = y_1 - a^2_{01} \quad \partial L / \partial a^2_{01} = -1$

$$\frac{\partial L}{\partial a^2_{01}} = \frac{\partial L}{\partial t} \times \frac{\partial t}{\partial a^2_{01}}$$

$$\frac{\partial t}{\partial a^2_{01}} = \frac{\partial y_1}{\partial a^2_{01}} - \frac{\partial a^2_{01}}{\partial a^2_{01}} = 0 - 1 = -1$$

$$= 2t \times (-1)$$

$$\frac{\partial L}{\partial a^2_{01}} = -2t = -2(y_1 - \hat{y}_1)$$

$$\boxed{\frac{\partial L}{\partial a^2_{01}} = -2(y_1 - a^2_{01})} \quad -(1)$$

$$a^1_{01} = \text{lin act}(z^1_{01})$$

$$a^2_{01} = z^2_{01}$$

$$\boxed{\frac{\partial a^2_{01}}{\partial z^2_{01}} = \frac{\partial z^2_{01}}{\partial z^2_{01}} = 1} \quad \text{— (2)}$$

$$z^2_{01} = w^2_{00}\, a^1_{01} + w^2_{01}\, a^1_{11}$$

$$\frac{\partial z^2_{01}}{\partial w^2_{01}} = 0 + \frac{\partial w^2_{01}\, a^1_{11}}{\partial w^2_{01}}$$

$$\boxed{\frac{\partial z^2_{01}}{\partial w^2_{01}} = a^1_{11}} \quad \text{— (3)}$$

$$\frac{\partial L}{\partial w^2_{01}} = \frac{\partial L}{\partial a^2_{01}} \times \frac{\partial a^2_{01}}{\partial z^2_{01}} \times \frac{\partial z^2_{01}}{\partial w^2_{01}}$$

$$\boxed{\frac{\partial L}{\partial w^2_{01}} = -2(y_1 - a^2_{01}) \times 1 \times a^1_{11}}$$

$$\boxed{\frac{\partial L}{\partial w^2_{00}} = -2(y_1 - a^2_{01}) \times 1 \times a^1_{01}}$$

$$w^2_{01(new)} = w^2_{01(old)} - \alpha \frac{\partial L}{\partial w^2_{01(old)}}$$

Similarly

$$w^2_{00(new)} = w^2_{00(old)} - \alpha \frac{\partial L}{\partial w^2_{00(old)}}$$

lets pick w^1_{12}

We need to find $\partial L / \partial w^1_{12}$

$L : (y_1 - \hat{y}_1)^2$

$L : (y_1 - a^2_{01})^2$

$a^2_{01} : \text{lin act}(z^2_{01})$

$z^2_{01} : w^2_{00} a^1_{01} + w^2_{01} a^1_{11}$

$a^1_{01} : \text{relu}(z^1_{01})$

$a^1_{11} : \text{relu}(z^1_{11})$

$z^1_{01} : w^1_{00} x_{01} + w^1_{01} x_{11} + w^1_{02} x_{12}$

$z^1_{11} : w^1_{10} x_{01} + w^1_{11} x_{11} + w^1_{12} x_{12}$

$$\frac{\partial L}{\partial w^1_{12}} = \frac{\partial L}{\partial a^2_{01}} \times \frac{\partial a^2_{01}}{\partial z^2_{01}} \times \frac{\partial z^2_{01}}{\partial a^1_{11}} \times \frac{\partial a^1_{11}}{\partial z^1_{11}} \times \frac{\partial z^1_{11}}{\partial w^1_{12}}$$

$$\frac{\partial L}{\partial w^1_{12}} = -2(y_1 - a^2_{01}) \times 1 \times w^2_{01} \times \frac{\partial \text{relu}(z^1_{11})}{\partial z^1_{11}} \times x_{12}$$

$$\frac{\partial L}{\partial w^1_{12}} = -2(y_1 - a^2_{01}) \times w^2_{01} \times \frac{\partial \text{relu}(z^1_{11})}{\partial z^1_{11}} \times x_{12}$$

$$\frac{\partial\, relu(z^1_{11})}{\partial z^1_{11}}$$

Derivatives of activation functions:

1) Linear activation

$$y = f(x) = x$$

$$Lin\,act(z^1_{11}) = z^1_{11}$$

$$\frac{\partial\, Lin\,act(z^1_{11})}{\partial z^1_{11}} = \frac{\partial z^1_{11}}{\partial z^1_{11}} = 1$$

2) ReLU Activation

$$f(x) = x \text{ if } x \geq 0$$
$$= 0 \text{ if } x < 0$$

$$f'(x) = \partial x/\partial x \text{ if } x \geq 0 = 1 \text{ if } x \geq 0$$

$$= \partial 0/\partial x \text{ if } x < 0 = 0 \text{ if } x < 0$$

$$\frac{\partial\, relu(z^1_{11})}{\partial z^1_{11}} = 1 \text{ if } z^1_{11} \geq 0$$
$$= 0 \text{ if } z^1_{11} < 0$$

